

The JDRF tranSMART instance can be accessed at <https://transmart.mijdrfcoe.org/transmart/login/auth>

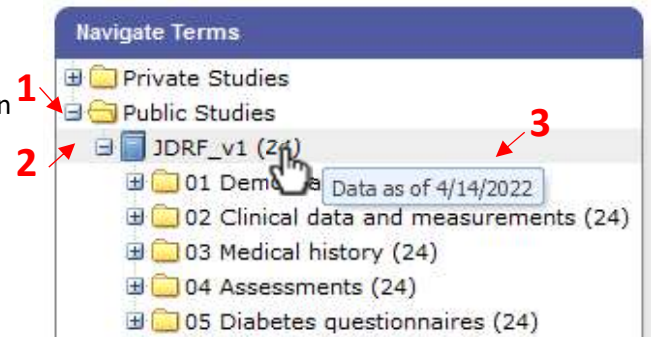
The application works best in Chrome, Firefox, or Safari. It cannot be used from a mobile device. The first time you log in (and once every 90 days after that), you will be asked to attest to a Data Use Statement.

Overview

To explore the data available, click on the plus sign (+) next to “Public Studies” in the **Navigate Concepts** section (1). Then click on the plus sign next to “JDRF_v1” to expand the terms tree (2).

The tree provides additional useful information:

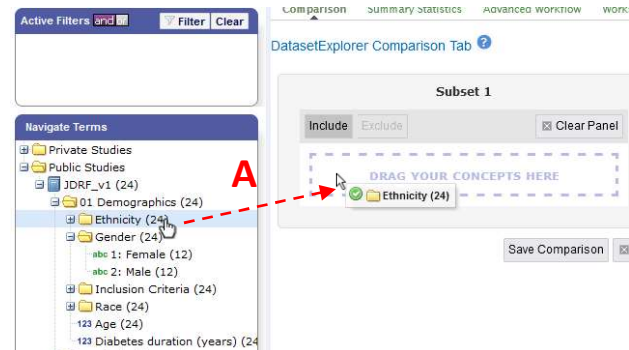
- The numbers in parentheses after the terms are the number of samples that have that data.
- The **123** or **abc** before a term indicates if the data is numeric or categorical in nature.
- Hovering over a node, or right-clicking on it, will bring up a definition pop-up for some terms (3).



Selecting Data

The first step is to set the **Subset(s)** of data you wish to explore. Drag and drop terms from the tree into the **Subset** boxes on the right (A).

- If an entire folder is dragged over, all samples with a value for that data point are included in the subset regardless of the value (e.g. dragging the folder “Ethnicity” would include all patients with any Ethnicity value).
- If only a single term is included, then only samples having that value are included (e.g. dragging “1: Female” would include only those patients who are Female).
- If a numeric term is selected, values can be further restricted by choosing an option from the pop-up menu. Select “No Value” to use all samples with any value.
- Terms in the same subset box are part of an “OR” statement.
- Subset boxes are joined together in “AND” statements.
- Two different subsets (**Subset 1** and **Subset 2**) can be defined and compared. Example B compares **Subset 1**: “1: Female” AND “Atorvastatin” OR “Simvastatin” to **Subset 2**: “1: Female” AND “0: No” Statin use.

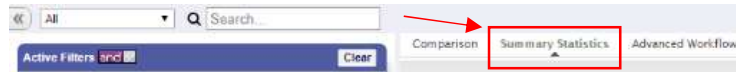


Analyzing Data

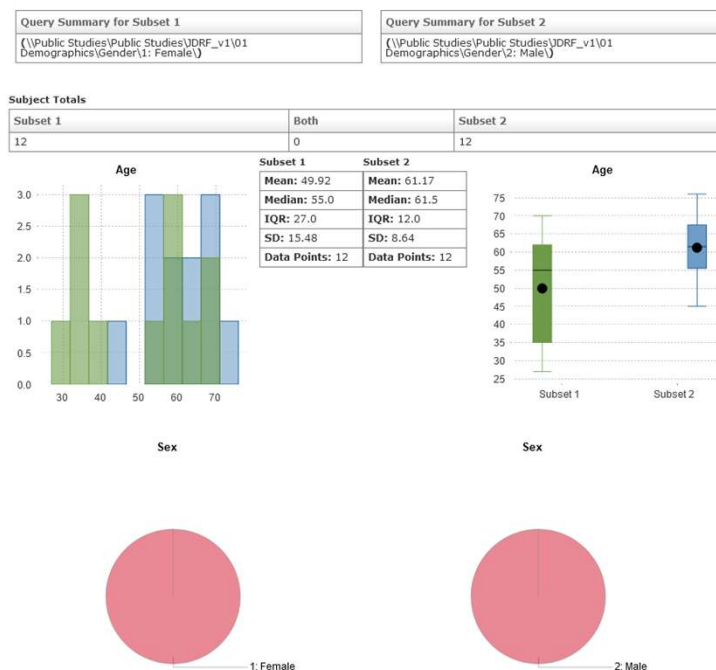
After setting a subset, clicking on the **Summary Statistics** tab will bring up an overview of the subset(s) by Age, Sex, and Race. Scroll through the results to see summaries and breakdowns of the terms used in the subset statements. For the subsets created, compare a concept across those Subsets by dragging an additional term from the tree directly onto the **Summary Statistics** results page.

Example: Compare eGFR between Female and Male in JDRF data.

- From the JDRF Navigate Terms tree, drag and drop “1: Female” into **Subset 1** and “2: Male” into **Subset 2**. Then click on **Summary Statistics**.



The top of the Summary Statistics screen shows a comparison of the two subsets:



- Drag and drop “eGFR, Creatinine-based formula (CKD-EPI 2021)(mL/min/1.73m2)” from the eGFR folder onto the Summary Statistics results page. A box plot comparing eGFR values in each subset will appear at the bottom of the page:

Analysis of eGFR, Creatinine-based formula (CKD-EPI 2021) (mL/min/1.73m2)

t-test demonstrated results are **not** significant at a 95% confidence level

p-value	2.151e-01
t-statistic	1.39594

